Arunabha Majumder

Mechatronics Engineer | Control Systems, Motion, System Modelling & Integration (MATLAB/Simulink)

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SUMMARY

Mechatronics engineer focused on control systems. Designs control strategies, models system dynamics and kinematics, and integrates sensors and actuators into prototypes tested at system level. Achieved 0.3–0.78% tracking error on a non-linear pneumatic actuator and a 25% energy reduction through a new actuation architecture. MATLAB/Simulink, Python.

KEY RESULTS

- **0.3–0.78%** tracking error with a gain-scheduled PID controller (I worked on the PID loop)
- **>25%** energy reduction from a new actuation system concept (Best Paper Award, 2026)
- **<10%** RMS error from a kinematic (Jacobian) force-estimation model

EXPERIENCE

Research Engineer – Mechatronic Systems & Control (PhD Project: VIEXO) | 2023 – Present
Aalborg University, Aalborg, Denmark

- Reduced user muscle effort by more than 15%, measured with EMG, by developing and implementing a control strategy (Assist-as-Needed hybrid control) on a mechatronic system, then testing and tuning it at system level.
- Cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026) by translating functional requirements into a system concept: a hybrid actuation architecture with a parallel spring and a 6 Nm motor.
- Achieved below 10% RMS error in end-effector force estimation, validated against a force sensor, by modelling the kinematics of a 5-bar planar parallel robot (Jacobian) and its variable-stiffness actuators.
- Selected and integrated sensors and actuators (motor, encoder, force sensors, DAQ, embedded control) into prototypes designed in SolidWorks and built through 3D printing.
- Solved system and integration problems by analysing measurement data from custom test rigs, working closely with electronics, control and biomechanics specialists; mentored Bachelor's students (2023 – 2025).

MTech Researcher – Control of Non-Linear Pneumatic Actuators (Master's Thesis) | 2020 – 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Brought position error down to 0.3-0.78%, better than a standard PID, by working on the outer PID loop of a neural-network gain-scheduled controller in MATLAB/Simulink. I collected the experimental data, fed it to the lab's existing neural network and used its output to set the PID gains (IEEE CONECCT 2022).
- Collected load-displacement data by fabricating pneumatic muscles to an existing lab design and testing them on the lab's setup (valves, pressure sensors, DAQ).

Bachelor's Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019
Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, machining, hands-on assembly and user testing; Best Major Project Award and INR 100,000 TATA Technologies Innovation Grant.

SKILLS

- **Control engineering:** PID tuning, gain scheduling, Assist-as-Needed control, motion control
- **Modelling & analysis:** System dynamics, kinematics, Jacobian-based estimation, analysis of test data

- **Tools:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino
- **Sensors & actuators:** Motors, encoders, force and pressure sensors, pneumatic valves, DAQ, embedded control
- **Prototyping:** SolidWorks (CSWA), 3D printing, test rigs
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022
AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019
Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer.

REFERENCES

Available upon request.